



深圳大夏龙雀科技有限公司
Shenzhen DX-SMART Technology Co Ltd.

DX-BT19-S 蓝牙模块
DX-BT19-S Bluetooth Module

Note: English instructions go to page 16
(英文说明书请跳转到第16页)

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手
册 v 2.0



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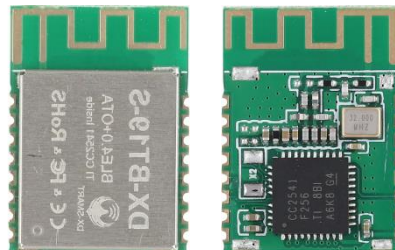
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一. 概述

DX-BT19-S 4.0蓝牙模块是深圳大夏龙雀科技有限公司专为智能无线数据传输而打造,采用美国TI公司CC2541芯片,配置 256Kb 空间,遵循V4.0 BLE蓝牙规范。支持AT 指令,用户可根据需要更改串口波特率、设备名称等参数,使用灵活。

本模块支持 UART 接口,并支持蓝牙串口透传,具有成本低、体积小、功耗低、收发灵敏性高等优点,只需配备少许的外围元件就能实现其强大功能。



二. 模块默认参数:

蓝牙协议	Bluetooth Specification V4.0 BLE
工作频率	2.4GHz ISM band
通信接口	UART
供电电源	3.3V
通信距离	30-40M (空旷环境)
外观尺寸	16.5(L)mm x 14.2(W)mm x 2(H) mm
蓝牙认证	FCC CE ROHS REACH
蓝牙名称	BT19-S
串口参数	9600、8数据位、1停止位、无校验、无流控
空中升级	本模块支持OTA空中升级
Service UUID	FFE0
Notify & Write UUID	FFE1
Write UUID	FFE2
Work temperature	MIN:-20℃ - MAX:+70℃
定制需求	如有其它特殊功能要求,可以联系我司,对模块进行定制

三. 应用领域:

DX-BT19-S 模块同时支持 BT4.2 BLE 协议,可以同具备 BLE 蓝牙功能的 iOS 设备直接连接,支持后台程序常驻运行。主要用于短距离的数据无线传输领域。避免繁琐的线缆连接,能直接替代串口线。**BT19-S 模块成功应用领域:**

- ※ 蓝牙无线数据传输;
- ※ 手持 POS 设备;
- ※ 手机、电脑周边设备;
- ※ 医疗设备无线数据传输;



- ※ 智能家居控制;
- ※ 蓝牙遥控玩具;
- ※ 蓝牙打印机;
- ※ 共享单车;

四. 功耗参数

模式	状态	电流	Unit
低功耗模式	Discoverable	400-700	uA
	Connected	8.5	mA
正常工作模式	Discoverable	8.5	mA
	Connected	8.5	mA

五. 射频特性

Rating	Value	Unit
BLE 发射功率	0	dBm
BLE 灵敏度	-93	dBm

六. 透传参数

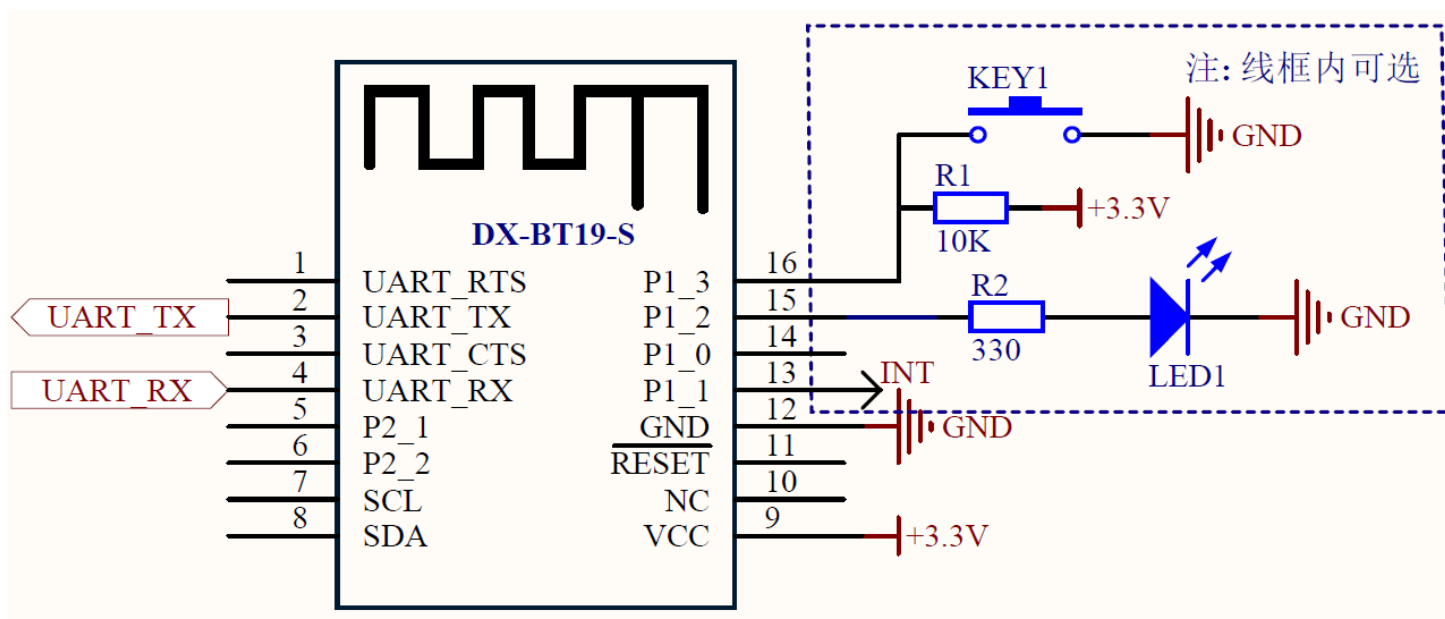
数据吞吐量:

Android -> BT19-S -> UART		UART -> BT19-S -> Android	
波特率	115200	波特率	115200
连接间隔时间(ms)	20	连接间隔时间(ms)	20
APP 数据包大小(bytes)	200	串口数据包大小(bytes)	200
发送间隔(ms)	70	发送间隔(ms)	70
吞吐量(bytes/s)	2500	吞吐量(bytes/s)	2800
Characteristic 写方式	Write without Response	Characteristic 通知方式	Notify
iPhone -> BT19-S -> UART		UART -> BT19-S -> iPhone	
波特率	115200	波特率	115200
连接间隔时间(ms)	30	连接间隔时间(ms)	30
APP 数据包大小(bytes)	100	串口数据包大小(bytes)	200
发送间隔(ms)	50	发送间隔(ms)	120
吞吐量(bytes/s)	2000	吞吐量(bytes/s)	1800
Characteristic 写方式	Write without Response	Characteristic 通知方式	Notify

注：此表格参数仅做参考，不代表模组能支持的最大数据吞吐量。



七. 模块引脚说明及最小电路图:



八. 管脚功能描述:

管脚序号	管脚名称	管脚说明
1	UART_RTS	悬空
2	UART_TX	串口数据输出
3	UART_CTS	悬空
4	UART_RX	串口数据输入
5	P2_1	调试数据口
6	P2_2	调试时钟口
7	SCL	时钟口
8	SDA	数据口
9	VCC	电源 V3.3
10	NC	悬空
11	RESETB	低电平复位, 至少5ms
12	GND	地
13	P1_1	蓝牙连接指示口 (未连接为低, 连接上为高)
14	P1_0	可编程输入输出口
15	P1_2	LED 灯管脚 (未连接闪烁, 连接常亮)
16	P1_3	断开连接引脚 (200ms 低电脉冲断开)

九. 功能引脚详细说明

1、P24 脚 (P1_2): LED 灯指示引脚

- 用于指示蓝牙模块所处状态，LED灯闪烁方式与蓝牙模块状态对应见下表：

模块	LED 显示	模块状态
从模块	均匀慢速闪烁 (800ms-on, 800ms-off)	待机状态
	长亮	连接状态
主模块	均匀闪烁 (300ms-on, 300ms-off)	搜索及连接中
	长亮	连接状态

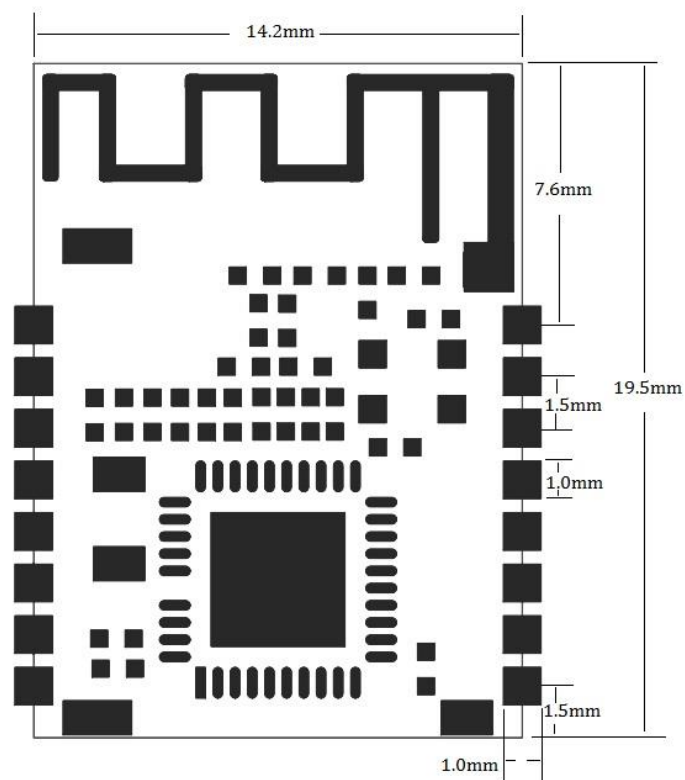
2、P25 脚 (P1_1): 连接状态指示脚

引脚状态	模块状态
输出低电平	待机状态
输出高电平	连接状态

3、P23 脚 (P1_3): 连接中断脚 (模块处于连接状态有效)

引脚状态	模块状态
无动作	连接状态
从模块输入 200ms 低电平脉冲	中断连接，模块进入待机状态

十. 外形尺寸:



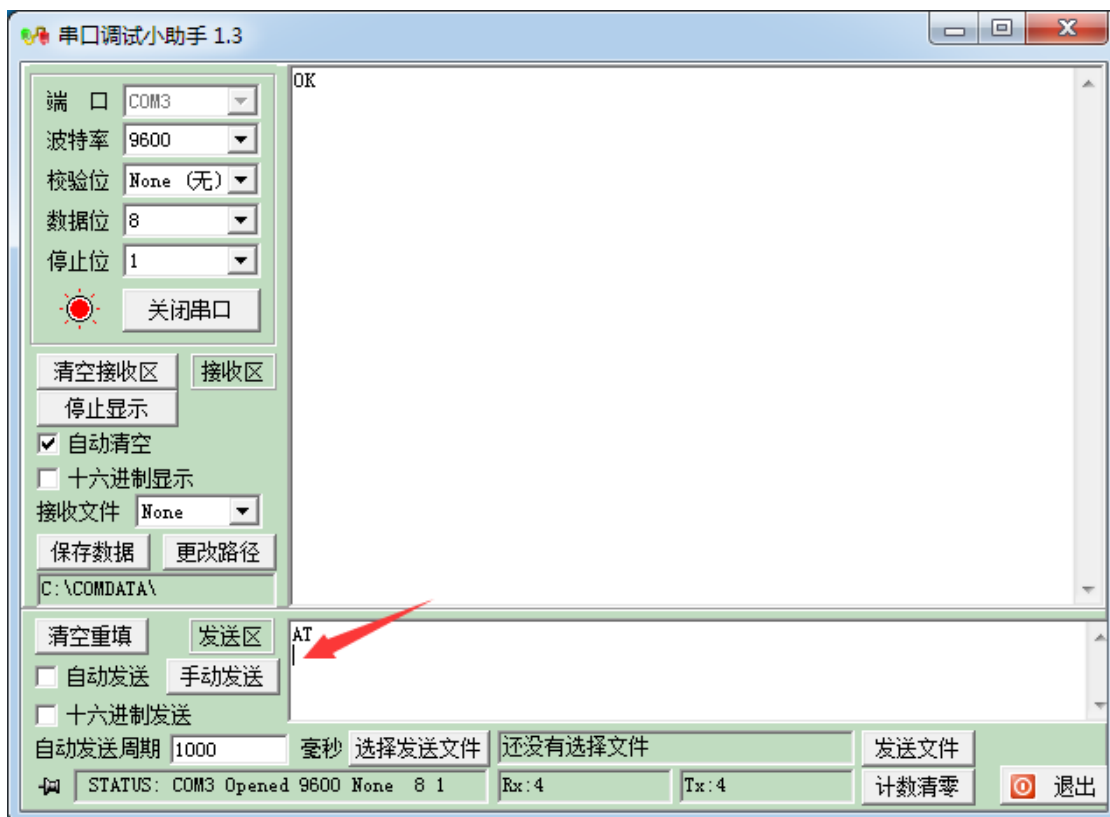
十一. LAYOUT 注意事项

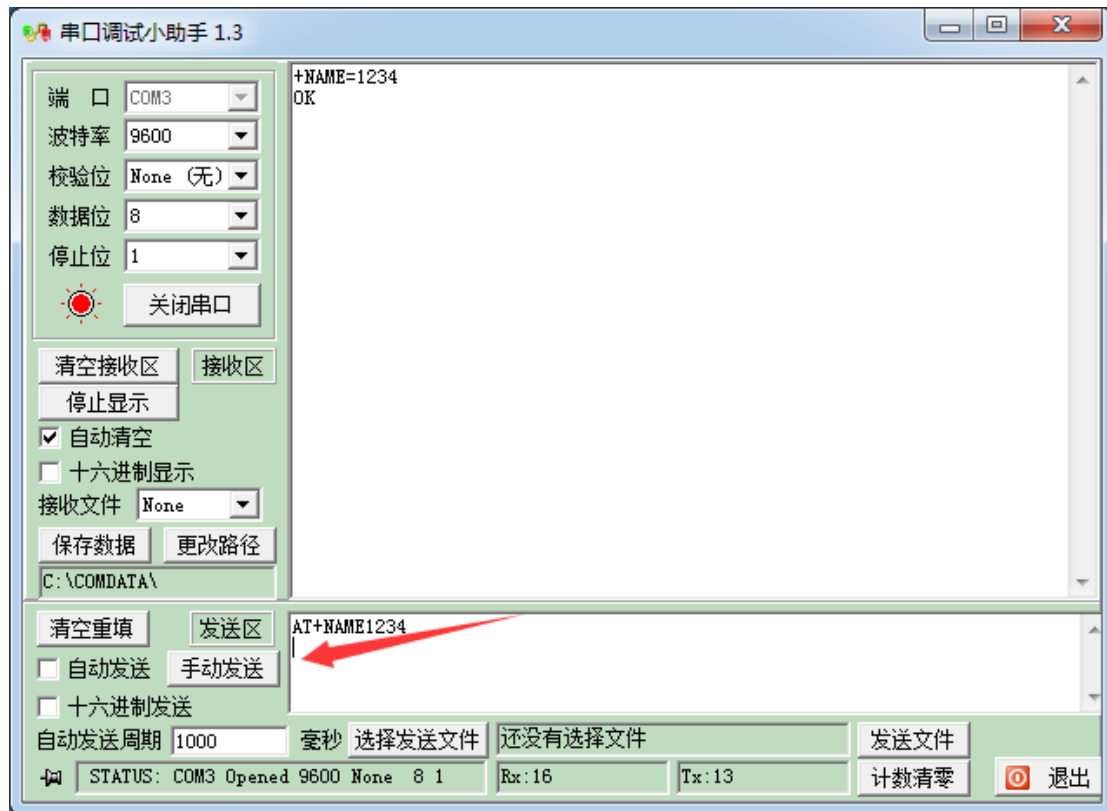
DX-BT19-S 蓝牙模块工作在2.4G无线频段，应尽量避免各种因素对无线收发的影响，注意以下几点：

- 1、包围蓝牙的产品外壳避免使用金属，当使用部分金属外壳时，应尽量让模块天线部分远离金属部分。产品内部金属连接线或者金属螺钉，应尽量远离模块天线部分。
- 2、模块天线部分应靠载板PCB 四围放置，不允许放置于板中，且天线下方载板铣空，与天线平行的方向，不允许铺铜或走线、或直接把天线部分直接露出载板。
- 3、建议在基板上的模块贴装位置使用绝缘材料进行隔离，例如在该位置放一个整块的丝印（TopOverLay）

十二. AT 指令集 (注：模块未连接时即为AT指令模式)

- 1、AT 指令，属于字符行指令，按行解析（即发 AT 指令时必须以回车换行或者\r\n、16 进制为 0D0A 结尾）
 - 2、AT 指令为大写，指令前缀为 AT+，可分为参数设置指令和读取指令。
 - 3、设置指令格式：AT+<CMD><PARAM>操作成功返回：+<CMD>=<PARAM>\r\n OK\r\n 失败不返回字符。
 - 4、读取指令格式：AT+<CMD>操作成功返回：+<CMD>=<PARAM>\r\n 失败不返回字符。
- AT 命令格式举例(图一为 AT 测试命令，图二为将蓝牙名称改为 1234)：





1、测试指令：

功能	指令	响应	说明
测试指令	AT\r\n	OK\r\n	

2、获取软件版本号：

功能	指令	响应	说明
查询版本号	AT+VERSION\r\n	+VERSION=<version>\r\n	<version>软件版本号

注：依据不同的模块与定制需求，版本会有区别。

3、设置\查询模块蓝牙地址码：

功能	指令	响应	说明
查询模块 MAC 地址	AT+LADDR\r\n	+LADDR=<laddr>\r\n	<laddr>蓝牙 MAC 地址码

4、设置\查询设备名称：

功能	指令	响应	说明
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查询模块蓝牙名	AT+NAME\r\n	+NAME=<name>\r\n	<name>蓝牙名，最长为 18 个字节 默认名称：BT19-S
设置模块蓝牙名	AT+NAME<name>\r\n	+NAME=<name>\r\n OK	

示例：

1. 发送设置：

AT+NAMEDX-BT19-S\r\n ——设置模块设备名为：“DX-BT19-S”

返回：

+NAME=DX-BT19-S\r\n ——设置模块设备名为：“DX-BT19-S”成功
OK

2. 发送查询：

AT+NAME\r\n ——查询模块名

返回：

+NAME= DX-BT19-S\r\n ——返回模块设备名为：“DX-BT19-S”

5、设置\查询—串口波特率：

功能	指令	响应	说明
查询模块波特率	AT+BAUD\r\n	+BAUD=<baud>\r\n	<baud>波特率对应序号 1:1200 2:2400 3:4800 4:9600 5:19200 6:38400 7:57600 8:115200 默认值：4 (9600)
设置模块波特率	AT+BAUD<baud>\r\n	+BAUD<baud>\r\n OK\r\n	

注：模块设置波特率后需重新上电，启用新波特率进行数据通信和 AT 指令解析。

示例：设置串口波特率：38400

1. 发送设置：

AT+BAUD6\r\n

返回：

+BAUD=6\r\n
OK\r\n

2. 发送查询：

AT+BAUD\r\n

返回：



+BAUD=6\r\n

6、设置\查询一串口停止位：

功能	指令	响应	说明
查询模块串口停止位	AT+STOP\r\n	+STOP=<Param>\r\n	<Param>停止位 0-1 停止位 1-2 停止位 默认值：0
设置模块串口停止位	AT+STOP<Param>\r\n	+STOP=<Param>\r\n OK	

7、设置\查询一串口校验位：

功能	指令	响应	说明
查询模块串口校验位	AT+PARI\r\n	+PARI=<Param>\r\n	<Param>校验位 0-1 无校验 1-2 奇校验 2-2 偶校验 默认值：0
设置模块串口校验位	AT+PARI<Param>\r\n	+PARI=<Param>\r\n OK	

8、设置\查询一通知上位机连接状态 (仅从模块有效)：连接成功模块返回 OK+CONN

功能	指令	响应	说明
查询通知上位机连接 状态	AT+NOTI\r\n	+NOTI=<Param>\r\n	<Param>校验位 0-不通知 1-通知 默认值：0
设置通知上位机连接 状态	AT+NOTI<Param>\r\n	+NOTI<Param>\r\n OK	

9、设置\查询一通知连接含地址码 (仅从模块有效)：连接成功模块返回 OK+CONN0x112233445566

功能	指令	响应	说明
查询通知上位机连接 状态	AT+NOTP\r\n	+NOTP=<Param>\r\n	<Param>校验位 0-不通知 1-通知 默认值：0
设置通知上位机连接 状态	AT+NOTP<Param>\r\n	+NOTP<Param>\r\n OK	



10、设置\查询—服务 SERVICE UUID：（修改完之后要重启手机蓝牙，手机才生效）

功能	指令	响应	说明
查询模块服务 UUID	AT+UUID\r\n	+UUID=<service>\r\n	<service>服务 UUID 默认服务 UUID： FFE0
设置模块服务 UUID	AT+UUID<service>\r\n	+UUID=<service>\r\n OK	

示例：设置服务 UUID 为：FF00

1. 发送设置：

AT+UUID0xFF00\r\n

返回：

+UUID=0xFF00 r\n

OK

11、设置\查询—通知 NOTIFY UUID\写入 WRITE UUID ：（修改完之后要重启手机蓝牙，手机才生效）

功能	指令	响应	说明
查询模块通知\写入 UUID	AT+CHAR\r\n	+CHAR=<UUID >\r\n	<UUID>通知\写入 UUID 默认值： FFE1
设置模块通知\写入 UUID	AT+CHAR<UUID>\r\n	+CHAR =<UUID>\r\n OK	

注：此通道是为可读写通道（即可以读也可写）

示例：设置通知\写入 UUID 为：FE01

1. 发送设置：

AT+CHAR0xFE01\r\n

返回：

+ CHAR =FE01r\n

OK\r\n

12、设置\查询—写入 WRITE UUID ：（修改完之后要重启手机蓝牙，手机才生效）

功能	指令	响应	说明
查询模块写入 UUID	AT+WRITE\r\n	+WRITE=<UUID >\r\n	<UUID>写入 UUID 默认值： FFE2
设置模块写入 UUID	AT+WRITE<UUID>\r\n	+ WRITE=<UUID>\r\n OK	



13、查询/设置—是否连接后等待建立通道：（仅从模块有效）

功能	指令	响应	说明
查询连接后等待建立通道	AT+WAIT\r\n	+ WAIT=<WAITE>\r\n	< WAITE>: 0: 不等待 1: 等待 默认: 0
设置连接后等待建立通道	AT+WAIT<WAITE> \r\n	+ WAIT=<WAITE>\r\n OK\r\n	

14、设置\查询—低功耗模式：（仅从模块有效）

功能	指令	响应	说明
查询模块低功耗模式	AT+PWRM\r\n	+PWRM=<Param>\r\n	< Param > (0、1) 0: 低功耗模式 1: 正常工作模式 默认值: 1
设置模块低功耗模式	AT+PWRM<Param>\r\n	+PWRM=<Param>\r\n OK	

15、设置/查询—广播时间间隔：（仅从模块有效）

功能	指令	响应	说明
查询模块广播时间间隔	AT+ ADVI \r\n	+ ADVI =<Param>\r\n	Param: 0~F 0—100ms 1—152.5ms 2—211.25ms 3—318.75ms 4—417.5ms 5—546.25ms 6—760ms 7—852.5ms 8—1022.5ms 9—1285ms A—2000ms B—3000ms C—4000ms D—5000ms E—6000ms F—7000ms 默认设置: 0
设置模块广播时间间隔	AT+ADVI<Param>\r\n	+ ADVI =<Param>\r\n OK	

注：此指令可以用于降低功耗

16、查询/设置—模块发射功率：（仅从模块有效）

功能	指令	响应	说明
查询模块发射功率	AT+POWE\r\n	+POWE=<POWE>\r\n	<POWE>:



设置模块发射功率	AT+POWE<POWE> \r\n	+POWE=<POWE>\r\n OK\r\n	0: -23dB 1: -6Db 2: 0dB 默认: 2
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17、软件重启:

功能	指令	响应	说明
软件重启	AT+RESET\r\n	OK\r\n	

18、恢复出厂设置:

功能	指令	响应	说明
恢复出厂设置	AT+DEFAULT\r\n	OK\r\n	

19、查询一主模块:

功能	指令	响应	说明
查询模块主模块	AT+ROLE\r\n	+ROLE=1\r\n	1: 主模块

注: 主模块只能搜索连接我司的蓝牙模块, 无法搜索连接其他的蓝牙模块设备。

20、手动搜索蓝牙设备: (仅主模块有效)

功能	指令	响应	说明
搜索蓝牙设备	AT+INQ\r\n	OK\r\n	

示例:

发送搜索:

AT+INQ\r\n

返回:

OK\r\n

+INQS\r\n

—— 开始

+INQ:1 0x001583000001 -63\r\n

—— 蓝牙设备 1

+INQ:2 0x001583000002 -56\r\n

—— 蓝牙设备 2

...

...

+INQE\r\n

—— 结束

Devices Found x

(x 代表数量)

21、打印搜索到设备列表: (仅主模块有效)

功能	指令	响应	说明
搜索蓝牙设备	AT+SHOW\r\n	搜到的设备列表	



22、设置\查询—自动搜索蓝牙设备：（仅主模块有效）

功能	指令	响应	说明
查询模块搜索模式	AT+AUTOINQ\r\n	+AUTOINQ=<Param> \r\n	< Param > (0、1) 0：手动搜索
设置模块搜索模式	AT+AUTOINQ<Param> \r\n	+AUTOINQ<Param>\r\n OK	1：自动搜索 默认值：0

23、手动连接蓝牙设备：（仅主模块有效）

功能	指令	响应	说明
连接蓝牙设备	AT+CONN<Param>\r\n	连接信息	Param：搜索的 1~9 设备序号

示例（若搜索到设备 1 ：0x001583000001）：

发送连接：

AT+CONN1\r\n

—— 连接序号为 1 的设备

返回：

+Connecting>>0x001583000001\r\n

—— 连接中

+Connected>>0x001583000001\r\n

—— 已连接

24、设置\查询—自动连接蓝牙设备：（仅主模块有效）

功能	指令	响应	说明
查询模块连接模式	AT+AUTOCONN\r\n	+AUTOCONN=<Param>\r\n	<Param> (0、1) 0：手动连接
设置模块连接模式	AT+AUTOCONN<Param> >\r\n	+AUTOCONN<Param>\r\n OK	1：自动连接 默认值：0

注：开启自动搜索自动连接后，模块会自动连接周边信号最强的模块。

25、连接远端指定地址蓝牙：（仅主模块有效）

功能	指令	响应	说明
连接蓝牙设备	AT+CONA<Param>\r\n	连接信息	Param: MAC 地址 如：0x112233445566

26、绑定指定地址蓝牙：（掉电后会记忆已绑定地址）（仅主模块有效）

功能	指令	响应	说明
连接蓝牙设备	AT+BIND<Param>\r\n	连接信息	Param: MAC 地址 如：0x112233445566



注：此指令需要先配置为自动搜索自动连接才生效。

27、清除连接记忆和清除已绑定：（仅主模块有效）

功能	指令	响应	说明
清除记忆	AT+CLEAR \r\n	OK\r\n	

注：AT+CLEAR 是用于清除主模块清除已绑定和记忆的最后一个从模块（主模块当设定为自动搜索自动连接时，连接了一个从模块，断开之后，将会继续寻找连接这个从模块，如果需要连接新的从模块，需要清除之前的记忆）。

十三．联系我们

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1. Overview

DX-BT19 4.0 Bluetooth module is built by Shenzhen DX-SMART Technology Co., Ltd. for intelligent wireless data transmission. It adopts CC2541 chip of American TI Company, configures 256Kb space, and follows V4.0 BLE Bluetooth specification. Support AT command, users can change the serial port baud rate, device name, pairing password and other parameters as needed, flexible use.

This module supports UART interface and supports Bluetooth serial port transparent transmission. It has the advantages of low cost, small size, low power consumption, high sensitivity of sending and receiving, etc. It can realize its powerful functions with only a few peripheral components simple operation, high cost performance and technology leading edge.

2. Module default parameters:

Bluetooth Protocol	Bluetooth Specification V4.0 BLE
Working Frequency	2.4GHz ISM band
Communication Interface	UART
Power Supply	3.3V
Communication distance	30-40M (Open and unobstructed environment)
Physical Dimension	18.5(L)mm x 13.5(W)mm x 2(H) mm
Bluetooth Authentication	FCC CE ROHS REACH
Bluetooth Name	BT19
Serial Port Parameters	9600、8 data bits、1 stop bit、No check、No flow control
Service UUID	FFE0
Notify\Write UUID	FFE1
Write UUID	FFE2
Storage temperature	MIN:-55℃ - MAX:+125℃
Work temperature	MIN:-20℃ - MAX:+70℃
Customized requirements	If you have other special function requirements, you can contact us to customize the module.



3. Application area:

DX-BT19 module supports BT4 .0 BLE protocol, which can be directly connected to iOS devices that have BLE Bluetooth function, and supports background program resident operation.

Successful application of BT19 module:

- ※ Bluetooth wireless data transmission;
- ※ Mobile phones, computer peripherals;
- ※ Handheld POS device;
- ※ Medical equipment wireless data transmission;
- ※ Smart Home Control;
- ※ Automotive Inspection OBD Equipment;
- ※ Bluetooth printer;
- ※ Bluetooth remote control toy;
- ※ Anti-lost device, LED light control;

4. Power consumption parameters:

Mode	Status	Current	Unit
Low power mode	Discoverable	400-700	uA
	Connected	8.5	mA
Normal working mode	Discoverable	8.5	mA
	Connected	8.5	mA

5. Radio frequency characteristics:

Rating	Value	Unit
BLE Transmit power	0	dBm
BLE Sensitivity	-93	dBm

6. Transparent transmission parameters

Data throughput:

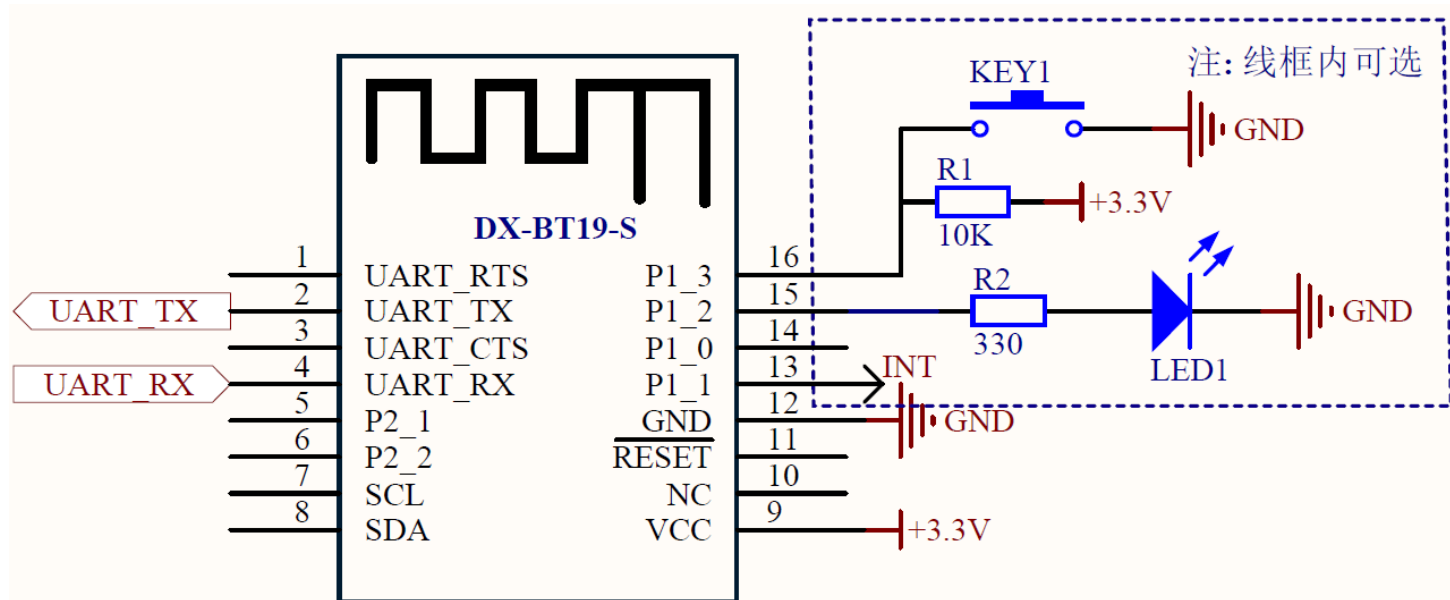
Android ->BT19 -> UART		UART ->BT19 -> Android	
Baud rate	115200	Baud rate	115200
Connection interval (ms)	20	Connection interval (ms)	20
Serial packet size (bytes)	200	Serial packet size (bytes)	200



Transmission interval (ms)	70	Transmission interval (ms)	70
Throughput (bytes/s)	2500	Throughput (bytes/s)	2800
Characteristic Write	Write without Response	Characteristic Notify	Notify
iPhone 6 -> BT19 -> UART		UART -> BT19 -> iPhone 6	
Baud rate	115200	Baud rate	115200
Connection interval (ms)	30	Connection interval (ms)	30
Serial packet size (bytes)	100	Serial packet size (bytes)	200
Transmission interval (ms)	50	Transmission interval (ms)	120
Throughput (bytes/s)	2000	Throughput (bytes/s)	1800
Characteristic Write	Write without Response	Characteristic Notify	Notify

Note: This table parameter is for reference only and does not represent the maximum data throughput that the module can support.

7. Module pin description and minimum circuit diagram:



8. Pin function description:

Pin number	Pin name	Pin description
1	UART_RTS	NC



2	UART_TX	Serial data output
3	UART_CTS	NC
4	UART_RX	Serial data input
5	P2_1	Debug data port
6	P2_2	Debug clock port
7	SCL	Clock port
8	SDA	Data port
9	VCC	3.3 V
10	NC	NC
11	RESETB	Low level reset, at least 5ms
12	GND	Land
13	P1_1	Bluetooth connection indicator (not connected low, connection high)
14	P1_0	Programmable input and output port
15	P1_2	LED light pin
16	P1_3	SW1 system button,

9. Detailed description of function pins:

1. P15 pin (P1_2): LED indicator pin

- Used to indicate the status of the Bluetooth module. Correspondence between the LED flashing mode and the Bluetooth module status is shown in the following table:

Mode	LED Display	Module Status
Slave module	Uniformly slow flashing (800ms-on, 800ms-off)	standby mode
	Long bright	Connection Status
Main module	Evenly flashing (300ms-on,300ms-off)	Search and connect
	Long bright	Connection Status

2. P13 pin (P1_1): connection status indicator

Pin state	Module status
Output low level	standby mode
Output high level	Connection Status

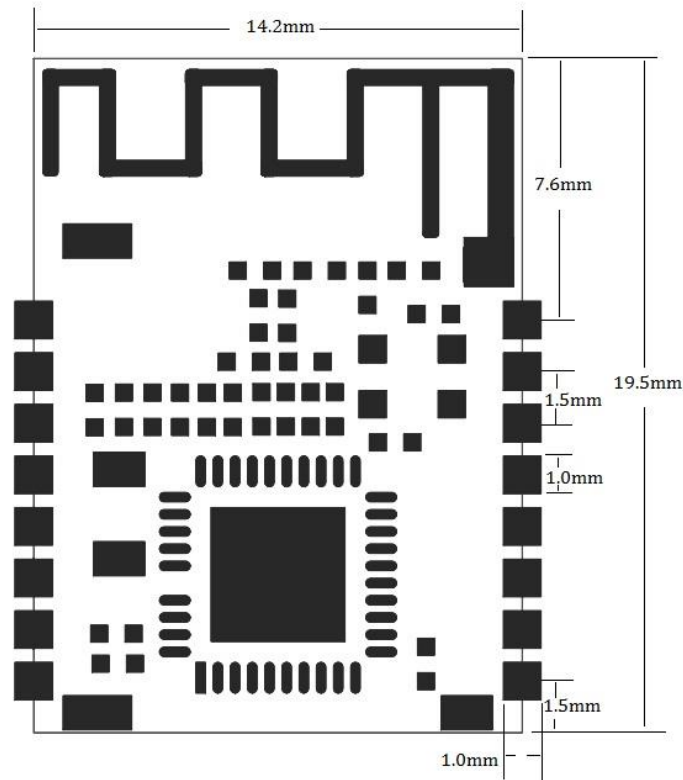
3. P16 pin (P1_3): connection interrupt pin (module is in the connected state)

Pin state	Module status
No action	Connection Status



Input 200ms low pulse	Interrupt connection, module enters standby
-----------------------	---

10. Dimensions:



11. LAYOUT Precautions:

The DX-BT19 Bluetooth module works in the 2.4G wireless band. It should try to avoid the influence of various factors on the wireless transceiver. Pay attention to the following points:

1. the product shell surrounding the Bluetooth module to avoid the use of metal, when using part of the metal shell, should try to make the module antenna part away from the metal part.
2. The internal metal connecting wires or metal screws of the product should be far away from the antenna part of the module.
3. The antenna part of the module should be placed around the PCB of the carrier board. It is not allowed to be placed in the board, and the carrier board under the antenna is slotted. The direction parallel to the antenna is not allowed to be copper or traced. It is also a good choice to directly expose the antenna part out of the carrier board.
4. It is recommended to use insulating material for isolation at the module mounting position on the substrate. For example, put a block of screen printing (TopOverLay) at this position.

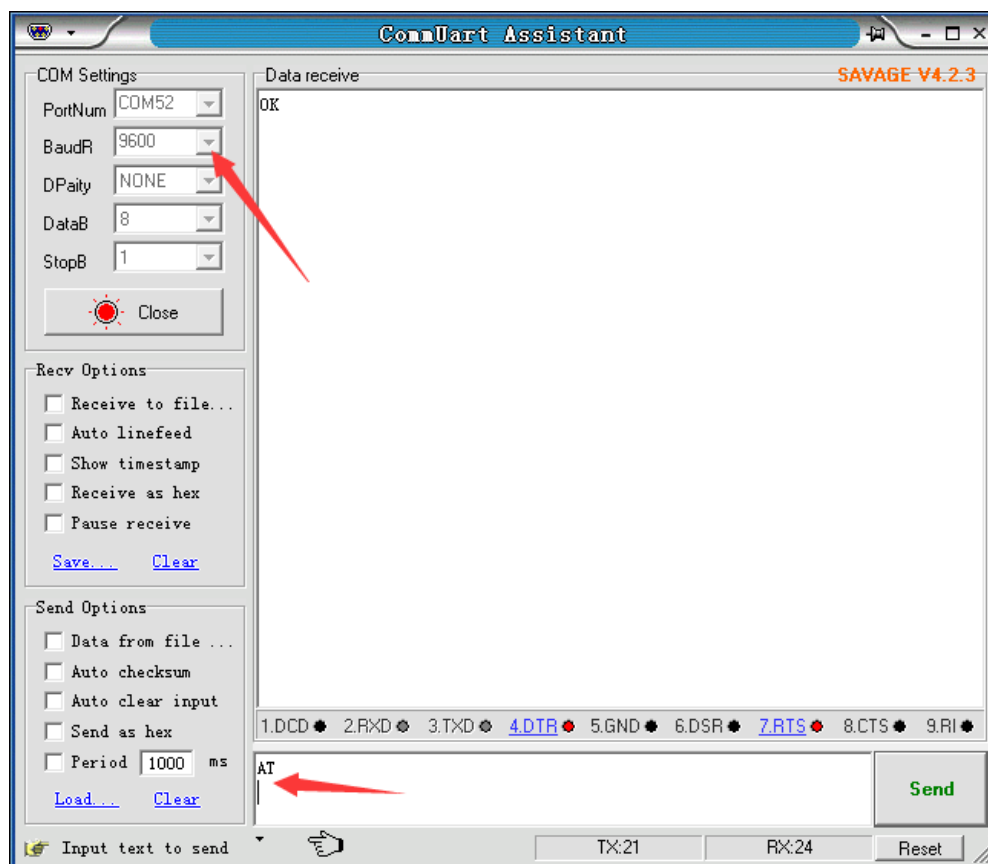


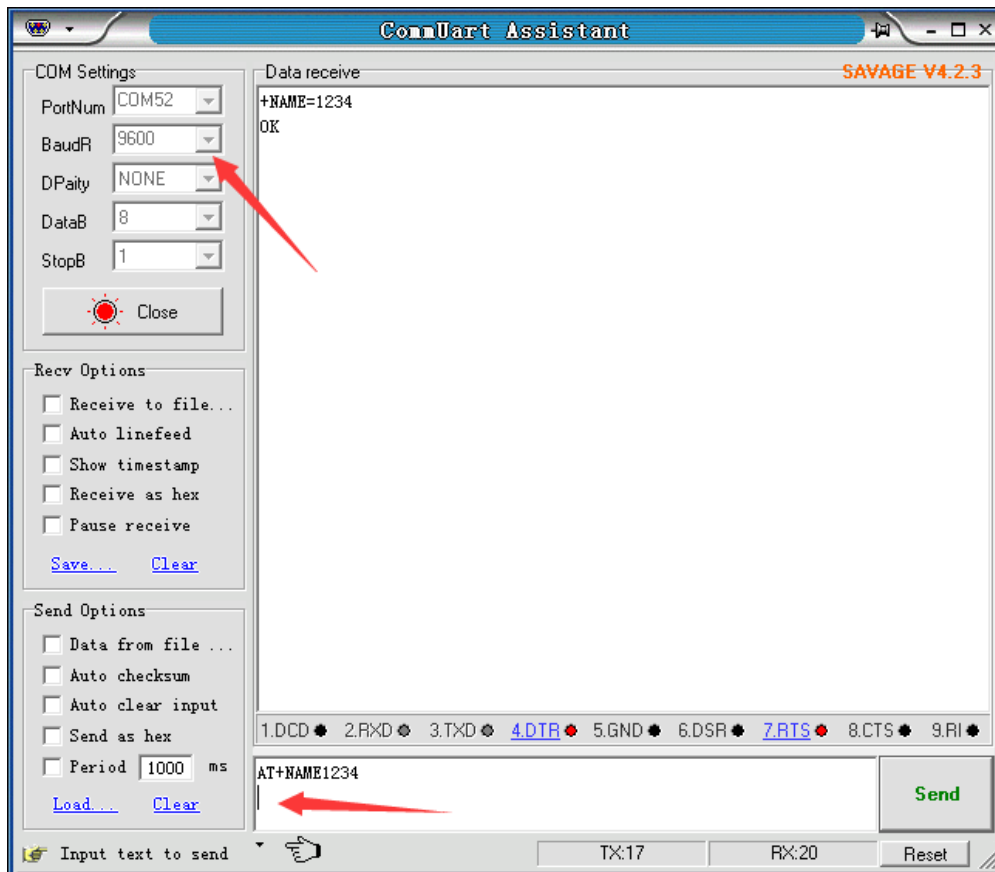
12. AT COMMAND

(Note: AT command mode when the module is not connected)

1. AT command, which belongs to the character line instruction, is parsed according to the line (that is, AT command must be returned by carriage return or \r\n, hexadecimal number is 0D0A)
2. The AT command supports case and the instruction prefix is AT+, which can be divided into parameter setting instructions and read instructions.
3. Set the instruction format: AT+<CMD><PARAM> Operation returns successfully: +<CMD>=<PARAM>\r\n OK\r\n Failure does not return characters.
4. Read instruction format: AT+<CMD>Operation succeeds: +<CMD>=<PARAM>\r\n Failure does not return a return character.

AT command format example (Figure 1 is AT test command, Figure 2 is to change the Bluetooth name to 1234):





1、Test Command:

Function	Command	Response	Description
Test instructions	AT \r\n	OK\r\n	

2、Get The Software Version:

Function	Command	Response	Description
Query version number	AT+VERSION\r\n	+VERSION=<version>\r\n OK\r\n	<version > Software version number

Note:The version will be different depending on different modules and customization requirements.

3、Set/Query Module Bluetooth MAC:

Function	Command	Response	Description
Query module MAC address	AT+LADDR\r\n	+LADDR=<laddr>\r\n	<laddr> Bluetooth 12-bit MAC Address Code

4、Set/Query Device Name:



Function	Command	Response	Description
Query module Bluetooth name	AT+NAME\r\n	+NAME=<name>\r\n	<name> Bluetooth name, up to 18 bytes Default name: BT19
Set the module Bluetooth name	AT+NAME<name>\r\n	+NAME=<name>\r\n OK	

Example:

1. Send Settings:

AT+NAME=DX-BT19\r\n ——Set module device name: “DX-BT19”

return:

+NAME=DX-BT19\r\n ——Set module device name: “DX-BT19” succeeded
OK\r\n

2. Send inquiry:

AT+NAME\r\n ——Query module name

return:

+NAME=DX-BT19\r\n ——Return module device name: “DX-BT19”

5、Set/Query - Serial Port Baud Rate:

Function	Command	Response	Description
Query module baud	AT+BAUD\r\n	+BAUD=<baud>\r\n	<baud> Baud rate corresponding serial number 1:1200 2:2400 3:4800 4:9600 5:19200 6:38400 7:57600 8:115200 Default: 4 (9600)
Set the module baud	AT+BAUD<baud>\r\n	+BAUD=<baud>\r\n OK\r\n	

Note: The module must be re-powered after setting the baud rate, enabling the new baud rate for data communication and AT command resolution.

Example: Setting the Serial Port Baud Rate: 38400

1. Send Settings:

AT+BAUD6\r\n

return:



+BAUD=6\r\n

OK\r\n

2. Send inquiry:

AT+BAUD?\r\n

return:

+BAUD=6\r\n

OK\r\n

6、Set/Query - Serial Port Stop Bit:

Function	Command	Response	Description
Query module serial port stop bit	AT+STOP\r\n	+STOP=<Param>\r\n	< Param> Stop bit 0 -1 Stop bit
Set module serial port stop bit	AT+STOP<Param>\r\n	+STOP=<Param>\r\n OK	1 -2 Stop bit Default: 0

7、Set / Query - Serial Parity Bit:

Function	Command	Response	Description
Query module serial parity bit	AT+PARI\r\n	+PARI=<Param>\r\n	< Param> Check Digit 0 -1 No check
Set the module serial parity bit	AT+PARI<Param>\r\n	+PARI=<Param>\r\n OK	1 -2 Odd parity 2 -2 Even parity Default: 0

8、Set/Query—Notify the host computer connection status (only slave module is valid): The connection success module returns OK+CONN:

Function	Command	Response	Description
Query status	AT+NOTI\r\n	+NOTI=<Param>\r\n	< Param> Check Digit 0- Not notified
Set status	AT+NOTI<Param>\r\n	+NOTI=<Param>\r\n OK	1- Notice Defaults: 0

9、Set/Query—Notification connection with address code (only slave module is valid): The connection success module returns OK+CONN0x112233445566:

Function	Command	Response	Description
Notification connection with address code	AT+NOTP\r\n	+NOTP=<Param>\r\n	< Param> Check Digit 0- Not notified



Notification connection with address code	AT+NOTP<Param m>\r\n	+NOTP=<Param>\r\n OK	1- Notice Defaults: 0
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10、Settings\Query—SERVICE UUID:

Function	Command	Response	Description
Query service UUID	AT+UUID\r\n	+UUID =<service>\r\n	<service> UUID
Set service UUID	AT+UUID<service >\r\n	+UUID =<service>\r\n OK	Default service UUID: FFE0

Example: Set the service UUID to: FE00

1. Send Settings:

AT+UUID0XFF00\r\n

return:

+UUID=0XFF00 r\n

OK

11、Settings\Query—NOTIFY UUID\ WRITE UUID:

Function	Command	Response	Description
Query module notify\write UUID	AT+CHAR\r\n	+CHAR=<UUID >\r\n	<UUID>notify\write UUID
Set module notify \write UUID	AT+CHAR<UUID> \r\n	+CHAR =<UUID>\r\n OK	Default: FFE1

Note: This channel is a readable and writable channel (ie it can be read or written)

Example: Set the notify \write UUID to: FE01

1. Send settings:

AT+CHAR0XFE01\r\n

return:

+CHAR= FE01r\n

OK\r\n

12、Settings\Query—WRITE UUID:

Function	Command	Response	Description
Query module write UUID	AT+WRITE\r\n	+WRITE=<UUID >\r\n	<UUID> write UUID Default: FFE2
Set module write UUID	AT+WRITE<UUID	+WRITE=<UUID>\r\n	



	>\r\n	OK	
--	-------	----	--

13、Settings\Query—Wait for the channel to be established after connecting:

Function	Command	Response	Description
Query	AT+WAIT\r\n	+WAIT=<wait>\r\n	<wait>
Set	AT+WAIT<wait>\r\n	+WAIT=<wait>\r\n OK	0: Not wait 1: wait Default: 0

14、Settings\Query - Low Power Mode: (only slave module is valid)

Function	Command	Response	Description
Query module low power mode	AT+PWRM\r\n	+PWRM=<Param>\r\n	< Param >(0、 1) 0: Low power mode
Set module low power mode	AT+PWRM<Param>\r\n	+PWRM=<Param>\r\n OK	1: working mode Default: 1

15、Settings\Query - Broadcast time interval: (only slave module is valid)

Function	Command	Response	Description
Query Broadcast time interval	AT+ ADVI \r\n	+ ADVI=<Param>\r\n	Param: 0~F 0—100ms 1—152.5ms
Set Broadcast time interval	AT+ADVI<Param>\r\n	+ ADVI=<Param>\r\n OK	2—211.25ms 3—318.75ms 4—417.5ms 5—546.25ms 6—760ms 7—852.5ms 8—1022.5ms 9—1285ms A—2000ms B—3000ms C—4000ms D—5000ms E—6000ms F—7000ms Default: 0

Note: This instruction can be used to reduce power consumption



16、Settings\Query - Module transmit power: (only slave module is valid)

Function	Command	Response	Description
Query module transmit power	AT+POWE\r\n	+POWE=<POWE>\r\n	<POWE>: 0: -23 dB 1: -6 dB 2: 0 dB Default: 2
Set module transmit power	AT+POWE<POWE>\r\n	+POWE=<POWE>\r\n OK\r\n	

17、Software restart:

Function	Command	Response	Description
Software restart	AT+RESET\r\n	OK\r\n	

18、Restore default settings:

Function	Command	Response	Description
Restore default settings	AT+DEFAULT\r\n	OK\r\n	

19、Query – Master Mode (only master module is valid):

Function	Command	Response	Description
Query module master module	AT+ROLE\r\n	+ROLE=<Param>\r\n	< Param > 1: master module

Note: The main module can only search for the Bluetooth module connected to our company, and cannot search for other Bluetooth module devices.

20、Search for Bluetooth devices (only master module is valid):

Function	Command	Response	Description
Search for Bluetooth devices	AT+INQ\r\n	OK\r\n	

Example:

Send search:

AT+INQ\r\n

return:

OK\r\n

+INQS\r\n

——Start

+INQ:1 0x001583000001 -63\r\n	——Bluetooth device 1
+INQ:2 0x001583000002 -56\r\n	——Bluetooth device 2
o o o	o o o
+INQE\r\n	——End
Devices Found x	(x represents the quantity)

21、 Print search to device list (only master module is valid):

Function	Command	Response	Description
Print list	AT+SHOW\r\n	Print search to device list	

22. Settings\Query—Automatically search for Bluetooth devices (only master module is valid):

Function	Command	Response	Description
Query module search mode	AT+AUTOINQ\r\n	+AUTOINQ=<Param>\r\n	< Param > (0、1) 0: Manual search
Set module search mode	AT+AUTOINQ<Param>\r\n	+AUTOINQ=<rParam>\r\n OK	1: Auto Search Defaults: 0

23、Connect a Bluetooth device (only master module is valid):

Function	Command	Response	Description
Connect a Bluetooth device	AT+CONN<Param>\r\n	Connection information	Param: 1~9 device serial number searched

Example (if searching for device 1 : 0x001583000001) :

Send connection:

AT+CONN1r\n — Connect the device with
sequence number 1

return:

```
+Connecting>>0x001583000001\r\n    ——connecting
+Connected>>0x001583000001\r\n    ——connected
```

24、Settings\Query—Automatically connect to Bluetooth device (only master module is valid):

Function	Command	Response	Description
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Query module connection mode	AT+AUTOCONN\r\n	+AUTOCONN=<Param>\r\n	< Param > (0、1) 0: Manual connection
Set module connection mode	AT+AUTOCONN<Param>\r\n	+AUTOCONN=<Param>\r\n OK	1: Auto connection Default: 0

25、Connect to the remote specified address Bluetooth (only master module is valid):

Function	Command	Response	Description
Connect a Bluetooth device	AT+CONA<Param>\r\n	Connection information	Param: MAC address Such as: 0x112233445566

26、Bind the specified address to Bluetooth: (The address will be remembered after power-off) (only master module is valid)

Function	Command	Response	Description
	AT+BIND<Param>\r\n		Param: MAC Example: 0x112233445566

Note: This command needs to be configured to automatically search for automatic connections to take effect.

27、Clear connection memory and clear bound: (only master module is valid):

Function	Command	Response	Description
Clear memory	AT+CLEAR \r\n	OK\r\n	

Note: AT+CLEAR is used to clear the last slave module of the main module to clear the bound and memorized (the main module is connected to a slave module when it is set to automatically search for automatic connection, after disconnecting, it will continue to find the connection. This slave module, if you need to connect a new slave module, needs to clear the previous memory).

13. Contact us

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